Los Angeles Crime Analysis Dashboard

Power BI Dashboard Project Report

INT 374

(DATA ANALYTICS WITH POWER BI)

(Project Semester Aug 2025-Jan 2026)

Submitted By:

Name: Yogesh

Reg. No.: 12307093

Roll No.: 9

Section: K23CN

Under the Guidance of

(Jaffar Amin Chacket Sir)

CERTIFICATE

This is to certify that Yogesh bearing Registration no. 12307093 has completed INT 374 project titled, "Los Angeles Crime Analysis Dashboard" under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature and Name of the Supervisor
Designation of the Supervisor
Department of Computer Science
Lovely Professional University
Phagwara, Punjab.

Date: 20:12:2025

DECLARATION

I, Yogesh, student of (Data Analytics with Power BI) under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 20:12:2025 Signature

Registration No. 12307093 **Yogesh**

ACKNOWLEDGEMENT

I would like to express my deepest appreciation to my supervisor, [Mr. Jaffar Amin Chacket Sir], for their expert guidance and encouragement throughout this project. I am also grateful to the faculty of [Department of CSE in Lovely Professional University] for providing the resources necessary to complete this work.

Special thanks go to my college friends for their unwavering support and patience. Lastly, I wish to thank my mentor [Mr. Gagandeep Kaur] for their valuable insights and data which were instrumental in the completion of this study.

Table of Content

Dataset Overview and Sources :

The dataset used for this project is sourced from Data.gov, specifically providing detailed incidents of crime in Los Angeles from 2020 to the present. It contains critical information such as the date and time of occurrence, geographic location (Area Name, Lat/Long), victim demographics (Age, Sex, Descent), and the type of crime committed.

- Dataset **Name:** Crime Data from 2020 to Present
- Source**:** catalog.data.gov - LA Crime Data

Data Preprocessing:

1. Data Cleaning (Power Query):

- **Loaded Raw Data:** Imported the Crime Data CSV file into Power BI.
- **Promoted Headers:** Ensured the first row was used as column names (e.g., Report ID, Date).
- **Changed Data Types:**
  - Converted **Date Rptd** and **DATE OCC** from text to **Date** format (so the Timeline Slicer works).
  - Converted **Latitude** and **Longitude** to **Decimal Number** (so the Map works).
  - Converted **Victim Age** to **Whole Number** (for the Scatter Plot).
- **Handled Missing Values:** Checked columns like Victim Sex and Weapon Desc to ensure blanks didn't break the visuals.

2. Data Modelling (Star Schema):

- **Created Dimension Tables:** You didn't just keep one big messy table. You split the data into focused tables to organize it:
  - **dim_area:** Stores location names (Area Name, Area Code).
  - **dim_status:** Stores case statuses (Status Desc, Status Code).
  - **dim_locationtype:** Stores premise details.
  - **dim_crime:** This is your main table containing the keys and values.
- **Built Relationships:** You connected these tables using **One-to-Many** relationships (e.g., connecting Area Code in the Area table to Area Code in the Crime table).

3. Measure Creation (DAX):

- **Calculated Key Metrics:** You created specific measures to count the data:
  - **Total Crimes:** Counts the number of Report IDs.
  - **Solved Rate %:** Calculates the percentage of cases where an arrest was made.
  - **Total Arrests:** Filters and counts only the solved cases.

4. Visualization Setup:

- **Map Configuration:** Set Latitude and Longitude fields specifically to "Don't Summarize" to show exact points.
- **Scatter Plot Logic:** Forced the scatter plot to show unique values by using Report ID in the details field.

Visualization and Analytics:

1. Page 1: The Crime Scene (Overview & Weapons)

This page focuses on the "What," "Where," and "Efficiency" of the police force.

- **KPI Cards (The Headlines):**
  - Total Crimes: A simple card displaying the raw count (558) to immediately show the scale of criminal activity.
  - Total Arrests: A card showing the number of successful apprehensions to contrast against the total crimes.
- **Efficiency Gauge (KPI Visual):**
  - Solved Rate %: A specialized visual that calculates the percentage of crimes resulting in an arrest. It is set against a 50% target to clearly show if the police are "Winning or Losing."
- **Crime Trends (Line Chart):**
  - Timeline Analysis: Plots Total Crimes on the Y-Axis and Year on the X-Axis. This visualizes the rise or fall of crime rates from 2020 to 2024.
- **Geospatial Analysis (Map):**
  - Hotspot Tracking: Uses Latitude and Longitude to plot exact crime locations.
  - Size Logic: The bubbles are sized by Total Crimes, making high-danger zones immediately visible on the map of Los Angeles.
- **Volume vs. Efficiency (Combo Chart):**
  - The Comparison: A "Line and Clustered Column Chart" that combines two different stories.
  - Bars: Show the Volume of crimes (e.g., Theft is high).
  - Line: Shows the Solved Rate (e.g., Theft has a low solved rate). This reveals which crimes are high-frequency but low-priority for arrests.
- **Weapon Lethality (Horizontal Bar Chart):**
  - The Tools: Ranks the top weapons (Handguns, Knives, etc.) by frequency.
  - Design: Uses a horizontal layout to make long text labels (like "Semi-Automatic Firearm") easy to read.

2. Page 2: The Victim Profile (Demographics)

This page shifts the focus to the "Who"—analysing the people affected by crime.

- **The "Starry Night" (Scatter Plot):**
  - Unique Analysis: The most complex chart in the report. It plots Average Victim Age vs. Average Incident Time.
  - The Hack: Uses Report ID in the "Values" field to force Power BI to render thousands of individual dots instead of summarizing them. This reveals hidden clusters of activity (e.g., 2 AM crimes targeting 25-year-olds).
- **Pulse of Danger (Area Chart):**

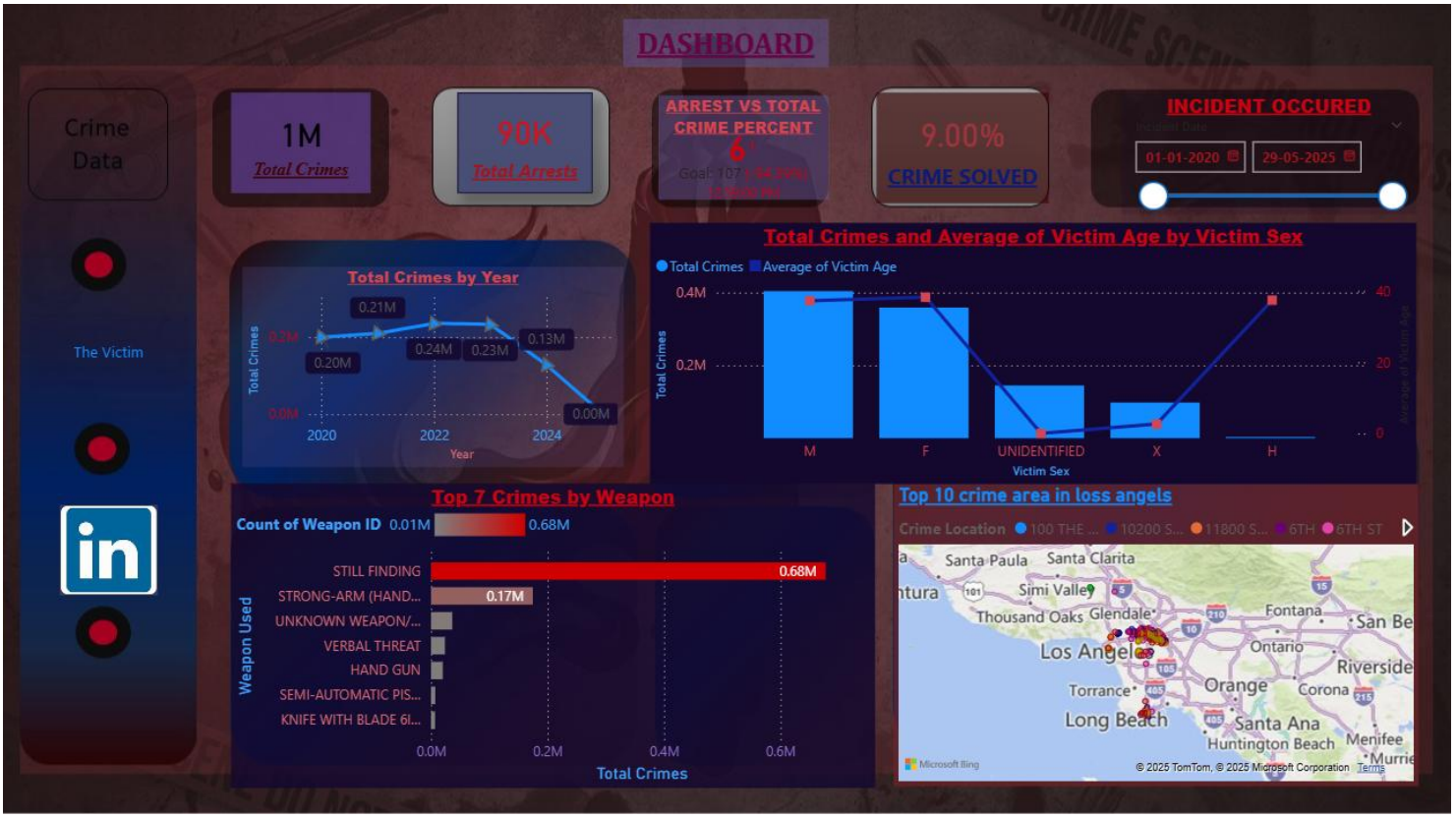
    - Age Distribution: Visualizes Victim Age on the X-Axis against Total Crimes on the Y-Axis.
    - The Shape: Creates a "Mountain" shape that peaks at the most vulnerable age group (20–35 years old).
- **Gender Split (Donut Chart):**
    - Ratio Analysis: A clear division showing the percentage of Male vs. Female victims, helping to identify gender-specific targeting patterns.
- **Vulnerability Matrix (Pivot Table):**
    - Heatmap: A cross-tabulation grid with Crime Type as rows and Victim Descent (Race) as columns.
    - Logic: Uses conditional formatting (Red backgrounds) to highlight specific demographic groups that are disproportionately targeted by specific crimes.
- **Demographic Risk (Clustered Bar Chart):**
    - Ranking: simple bar chart ranking Victim Descent groups by the volume of crimes they face.

3. UI/UX Engineering (The Noir Theme PPT image)

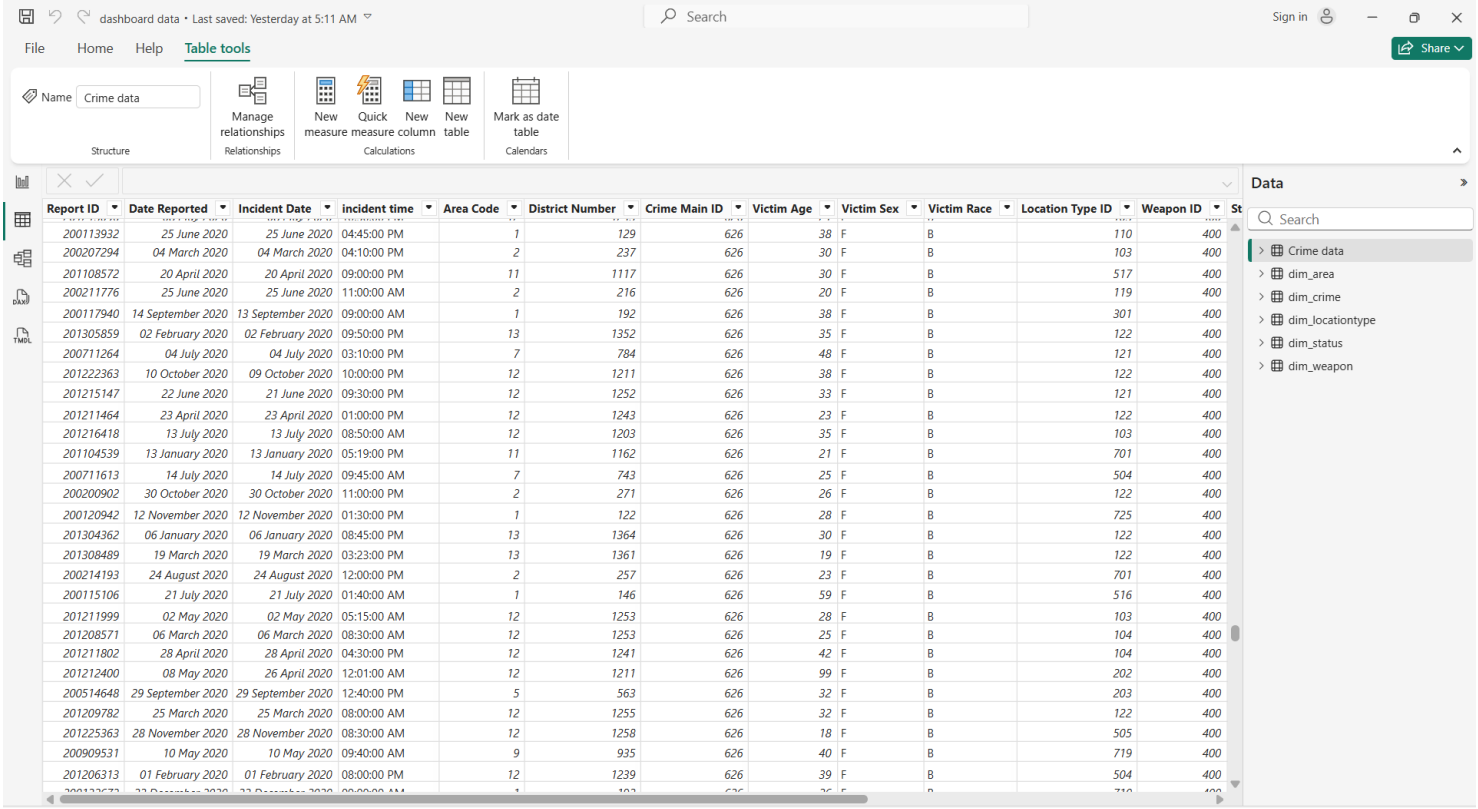
The design elements that make the dashboard feel like a professional app.

- **Invisible Navigation:**
    - Placed transparent buttons over the custom "Red Dots" on the background image.
    - Configured Page Navigation actions to allow users to switch seamlessly between the "Crime Map" and "Victim Data" pages.
- **External Connectivity:**
    - Added Web URL actions to the LinkedIn and GitHub icons, turning static images into active portfolio links.
- **The Noir Aesthetic:**
    - Applied a custom colour palette: Blood Red (#D00000) for alerts, Police Blue (#00509D) for standard data, and Glass Black (60% transparency) for backgrounds.
    - Removed standard white backgrounds from all visuals to ensure they blend perfectly into the detective-themed wallpaper.

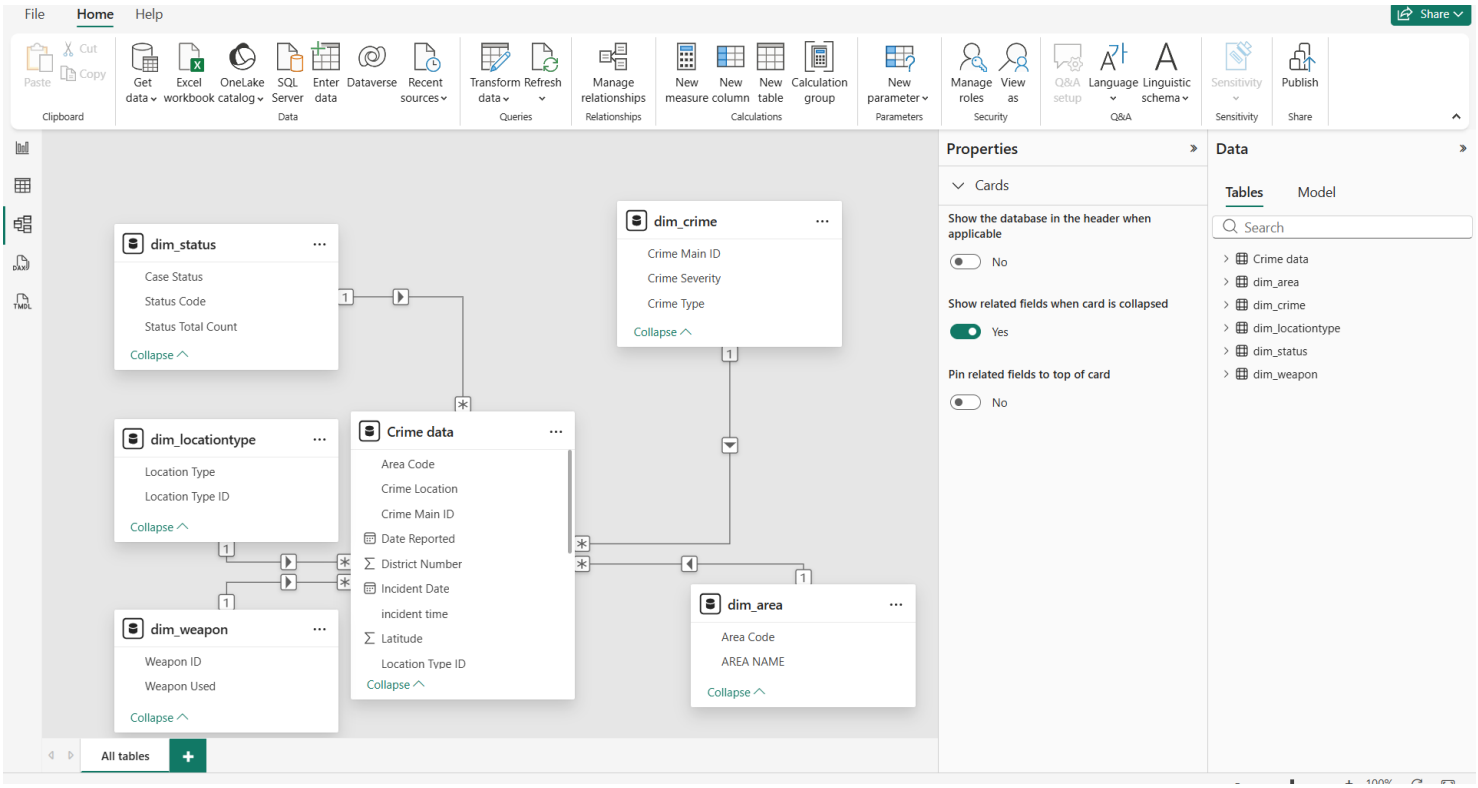
Screenshots



Crime Type	F	H	M	UNIDENTIFIED	X	Total
ARSON	42.60		32.04	0.00	1.62	25.65
ASSAULT WITH DEADLY WEAPON ON POLICE OFFICER	7.96	0.00	9.61		3.85	7.53
ASSAULT WITH DEADLY WEAPON, AGGRAVATED ASSAULT	36.85	33.67	37.14	17.07	13.91	36.62
ATTEMPTED ROBBERY	39.54	49.00	36.86	10.67	2.51	33.97
BATTERY - SIMPLE ASSAULT	39.45	40.14	40.68	0.00	20.39	39.97
BATTERY ON A FIREFIGHTER	24.00		28.59		3.03	25.00
BATTERY POLICE (SIMPLE)	14.89		12.83	9.50	3.32	8.59
BATTERY WITH SEXUAL CONTACT	30.88		29.85	0.00	11.55	30.69
BEASTIALITY, CRIME AGAINST NATURE SEXUAL ASSLT WITH ANIM	35.50		19.60		0.00	18.78
Total	38.29	37.46	37.31	0.01	2.62	28.92



Report ID	Date Reported	Incident Date	incident time	Area Code	District Number	Crime Main ID	Victim Age	Victim Sex	Victim Race	Location Type ID	Weapon ID
200113932	25 June 2020	25 June 2020	04:45:00 PM	1	129	626	38	F	B	110	400
200207294	04 March 2020	04 March 2020	04:10:00 PM	2	237	626	30	F	B	103	400
201108572	20 April 2020	20 April 2020	09:00:00 PM	11	1117	626	30	F	B	517	400
200211776	25 June 2020	25 June 2020	11:00:00 AM	2	216	626	20	F	B	119	400
200117940	14 September 2020	13 September 2020	09:00:00 AM	1	192	626	38	F	B	301	400
201305859	02 February 2020	02 February 2020	09:50:00 PM	13	1352	626	35	F	B	122	400
200711264	04 July 2020	04 July 2020	03:10:00 PM	7	784	626	48	F	B	121	400
201222363	10 October 2020	09 October 2020	10:00:00 PM	12	1211	626	38	F	B	122	400
201215147	22 June 2020	21 June 2020	09:30:00 PM	12	1252	626	33	F	B	121	400
201211464	23 April 2020	23 April 2020	01:00:00 PM	12	1243	626	23	F	B	122	400
201216418	13 July 2020	13 July 2020	08:50:00 AM	12	1203	626	35	F	B	103	400
201104539	13 January 2020	13 January 2020	05:19:00 PM	11	1162	626	21	F	B	701	400
200711613	14 July 2020	14 July 2020	09:45:00 AM	7	743	626	25	F	B	504	400
200200902	30 October 2020	30 October 2020	11:00:00 PM	2	271	626	26	F	B	122	400
200120942	12 November 2020	12 November 2020	01:30:00 PM	1	122	626	28	F	B	725	400
201304362	06 January 2020	06 January 2020	08:45:00 PM	13	1364	626	30	F	B	122	400
201308489	19 March 2020	19 March 2020	03:23:00 PM	13	1361	626	19	F	B	122	400
200214193	24 August 2020	24 August 2020	12:00:00 PM	2	257	626	23	F	B	701	400
200115106	21 July 2020	21 July 2020	01:40:00 AM	1	146	626	59	F	B	516	400
201211999	02 May 2020	02 May 2020	05:15:00 AM	12	1253	626	28	F	B	103	400
201208571	06 March 2020	06 March 2020	08:30:00 AM	12	1253	626	25	F	B	104	400
201211802	28 April 2020	28 April 2020	04:30:00 PM	12	1241	626	42	F	B	104	400
201212400	08 May 2020	26 April 2020	12:01:00 AM	12	1211	626	99	F	B	202	400
200514648	29 September 2020	29 September 2020	12:40:00 PM	5	563	626	32	F	B	203	400
201209782	25 March 2020	25 March 2020	08:00:00 AM	12	1255	626	32	F	B	122	400
201225363	28 November 2020	28 November 2020	08:30:00 AM	12	1258	626	18	F	B	505	400
200909531	10 May 2020	10 May 2020	09:40:00 AM	9	935	626	40	F	B	719	400
201206313	01 February 2020	01 February 2020	08:00:00 PM	12	1239	626	39	F	B	504	400



Links:

[*LinkedIn Post Link*]

[*GitHub Repository Link*]

Dashboard file LINK: [google drive]